

Digital Governance & Conflict

AI, Human Rights, and Relational Databases in Political Science

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Umbrella Research Framework

International Relations

My core research examines the intersection of **International Relations** and **AI adoption**. I analyze how technological transfers shift state behaviour and impact citizens' rights globally.

AI & Human Rights

Investigating the trade-offs between "security-augmented" technologies and **Physical Integrity Rights**, specifically the impact of autonomous digital systems on non-kinetic repression.

Addressing Research Bottlenecks



Siloed Data

Transfer logs and human rights indices are rarely connected, obscuring the causal link between tech and rights.



Static Reports

Traditional scholarship relies on static PDFs. We need dynamic tools to query real-time regime shifts.

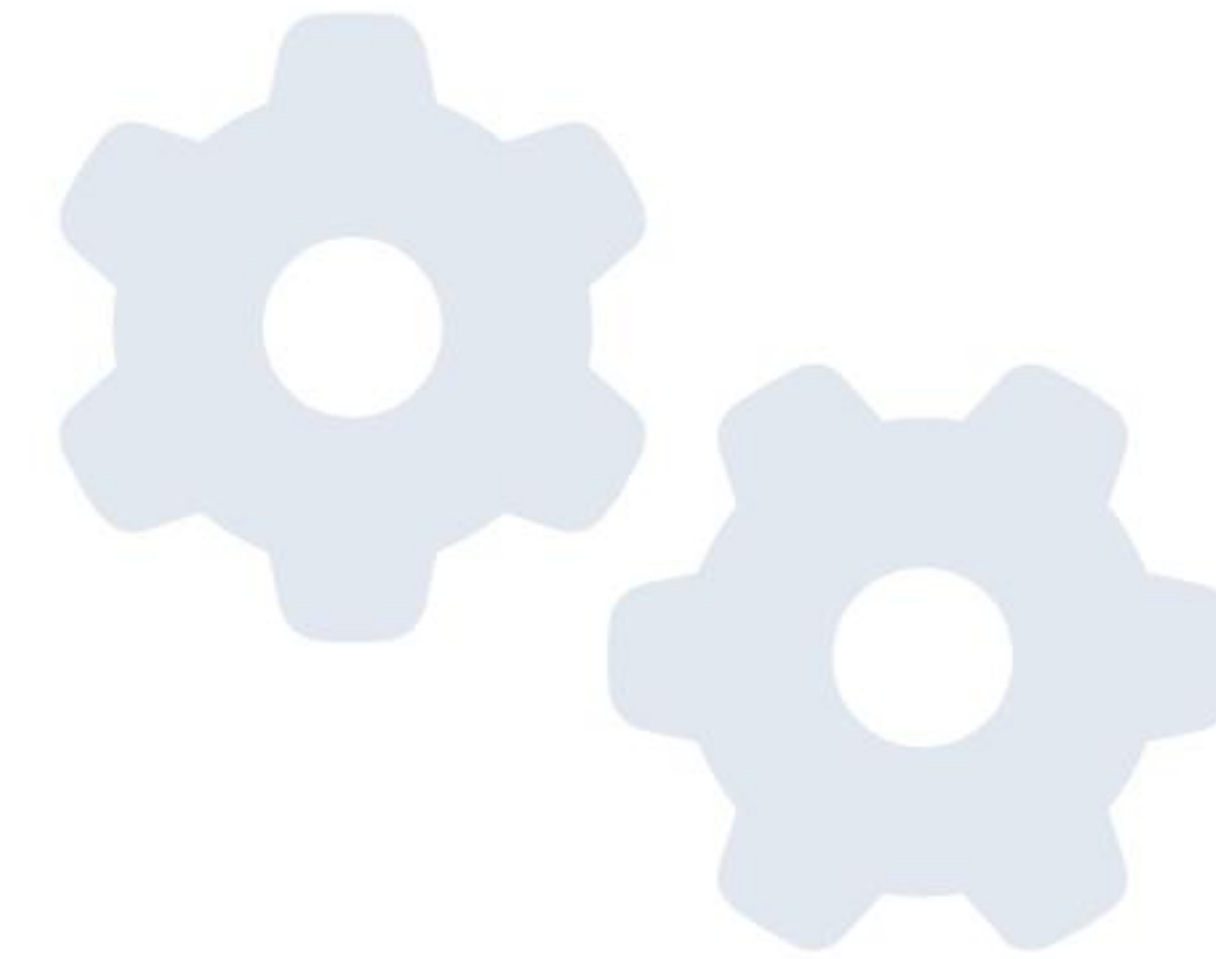


Relational Gaps

Spreadsheet-based data lacks the structural integrity required for complex longitudinal cross-national study.

End-to-End Technical Pipeline

- **PostgreSQL:** Serving as the "Truth" tier, housing structured relational data on countries and tech.
- **R & Tidyverse:** Performing the cleaning, merging, and reactive statistical computation.
- **Shiny:** Translating code into an interactive interface for policymakers.
- **Cloud Hosting:** Utilizing ShinyApps.io and GitHub for public dissemination of the scholarship.



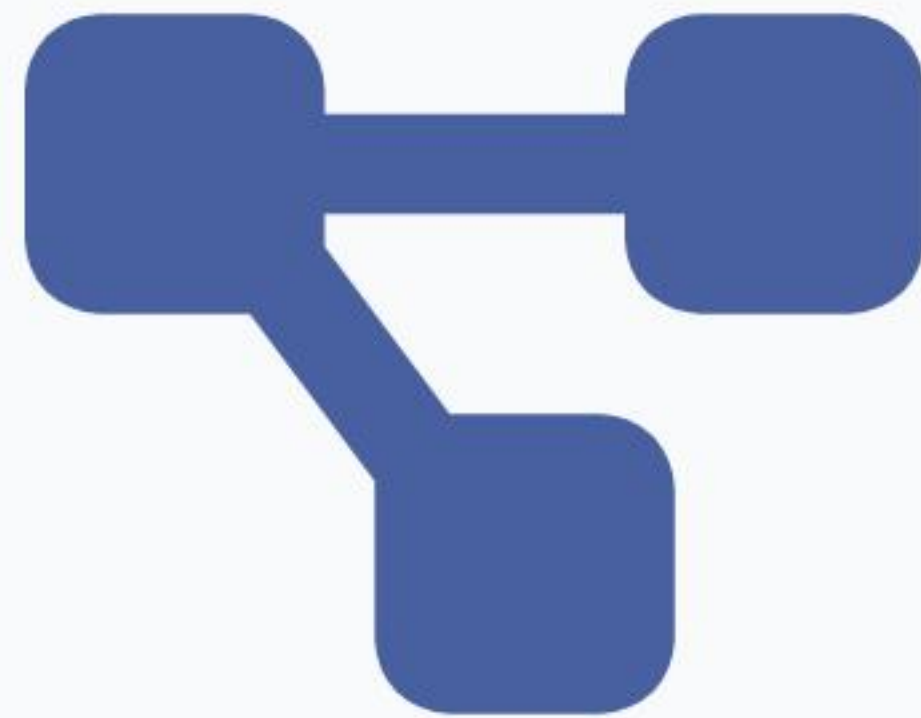
Relational Schema Structure

The database is normalized into four primary entities to ensure data integrity and query efficiency:

- **Transfers Table:** Tracks technology flow between nations (Exporters, Importers, Tech IDs).
- **AI Technologies Table:** Categorizes systems by risk levels and autonomy metrics.
- **Countries Table:** Metadata on regimes, regions, and levels of democracy.
- **Human Rights Table:** Longitudinal indexing of physical integrity and censorship metrics.



Entity Relationship Logic



Conceptual Data Mapping

Logical Data Flow

The system maps how abstract research concepts interact in a structured environment:

- **PK/FK Constraints:** Transfers are strictly bound to valid Country and Tech IDs.
- **Temporal Joins:** Human rights scores are linked to Country IDs by the specific year of transfer.
- **Consistency:** Ensures every visual point on the dashboard represents a valid relational record.

Quantitative Comparison Avg



Comparative Physical Integrity Index scores across regime types in our project's simulated dataset.

Advancing AI & Rights Research

This project contributes to the scholarship of AI and Human Rights by:

- **Operationalizing Hypotheses:** Directly testing the correlation between tech transfers and rights erosion.
- **Mapping Digital Repression:** Moving beyond kinetic force to see how AI surveillance affects physical integrity.
- **Tool for Policymakers:** Providing a rigorous methodology for tracking technological impact across regime types.



Research Utility

Three-Tier Server Architecture



Database Server

PostgreSQL: The "Memory."
Manages storage and ensures relational data integrity across all tables.



Application Server

R/Shiny: The "Brain."
Executes code, transforms data, and handles reactive user interface requests.



Web Server

ShinyApps.io: The "Interface."
Delivers the visual experience to the final user's browser via the cloud.

Project Outcome: The Dashboard



Digital Research Asset

By embedding this dashboard into my GitHub portfolio, I have created a permanent, interactive research asset that bridges the gap between data science and IR theory.

Professional Outlook & Impact

Employability & GitHub

This project serves as a key technical asset on my portfolio, showcasing professional skills in **SQL, R, and Cloud Deployment** as a high-value side-project.

CV & Dissertation

Demonstrates technical authority. This framework scales directly into the longitudinal data analysis for my **May 2027 dissertation**.

Questions?

Thank you for your attention.

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